

Charge Fibres and Taub Obstructions

Adjacency bridge note

Bridge note for mathematical relativity and linearization-stability researchers. Status: draft allowed, 17 July 2026.

1. Their object and the unresolved question

Linearization stability asks whether every solution of the linearized field equations is tangent to an actual family of exact solutions. On compact Cauchy surfaces with symmetries, quadratic Taub constraints can obstruct such an extension. The modern genericity result of Saraykar and Rai places this in the class of constant-mean-curvature spacetimes with compact Cauchy hypersurfaces ([arXiv:1609.07703](#)); it builds on the Fischer–Marsden–Moncrief tradition.

Our adjacent question is whether the obstruction depends on the **global charge fibre**: is a tangent obstructed on a fixed $U(1)$ bundle but extendible when electric or magnetic flux is admitted as part of the family?

	quadratic source	
$H^0_{\text{lin}} \times H^0_{\text{lin}}$	----->	coker L
fixed bundle: charge row unavailable		Taub pairing
enlarged flux family: new charge direction		may be absorbed
+-----+		+-----+

2. Exact dictionary

Item	Linearization-stability language	This project	Conversion or caveat
Background	Einstein or Einstein–matter solution	compact $\mathbb{R} \times S^1 \times S^2$	It is simultaneously a tuned Weyl–Maxwell background.
Linear tangent	kernel of the linearized equations modulo gauge	Einstein–Maxwell product standard photon/graviton harmonics and extra Weyl classes	The present bilinear is before the final residual quotient.
Quadratic constraint	Taub charge paired with a Killing initial datum	constant-lapse component of $D^2E[\Phi_1, \Phi_1]$	Other adjoint-cokernel classes remain to be classified.
Global sector	fixed conserved charges	fixed Chern class P_N , with electric variation tracked separately	A continuous magnetic variation changes the bundle family.
Extension	solution of $L\Phi_2 = -\frac{1}{2}D^2E[\Phi_1, \Phi_1]$	explicit correction or normalized adjoint witness	A removable output block is not proof that the real tangent extends.
Pairing	constraint/cokernel pairing	relative Taub form $\mathfrak{D}$	Exact algebraic arithmetic is used.

3. Reproduced benchmark

The compact topology fixes the magnetic sector. On the rational fixture the bundle number expands as

$$N(\epsilon) = 2 + 2\epsilon^2 p.$$

A smooth family on the same $N = 2$ bundle therefore has $p = 0$. The constant-lapse Taub component cannot be removed by silently varying magnetic flux. On the declared four-dimensional fixture span—radion, duality, $\ell = 1$ photon, and plus-branch $\ell = 2$ graviton—the symmetric form is

$$\text{diag}\left(-2, -\frac{1}{2}, -\frac{16}{3}, -\frac{72}{5} - 12\sqrt{3}\right).$$

Gauge descent and Cauchy-slice conservation follow from the coupled action Noether identities. The benchmark is independently replayed by:

```
python3 -m bridge.einstein_sector.compact_harmonic_domain_taub_descent \
--verify bridge/certificates/compact_harmonic_domain_taub_descent.json
python3 bridge/einstein_sector/verify_compact_harmonic_domain_taub_descent.py
```

4. Added result

Scoped charge-fibre obstruction. On the declared compact fixed bundle, several physical photon, graviton, and extra-Weyl tangents have nonzero second-order Taub pairings and therefore no smooth periodic second-order correction. Allowing a formal magnetic coefficient can remove specified constant-lapse components, but that coefficient belongs to an enlarged flux family, not the original fixed-bundle phase space.

The most recent real test uses the degenerate axial-polar $\ell = 2, k = 0$ minus-frequency Einstein pair. Its Hermitian Taub form is positive definite, so every nonzero real combination is obstructed at fixed bundle topology. A separate axial-polar sum-frequency source has an exact correction; it does not remove the zero-frequency constraint arising from conjugate self-products. Thus source solvability must be checked in every selection channel before declaring extension.

5. Consequence in their language

The result promotes “hold the charge fixed” from an informal side condition to part of the domain of the obstruction map. It suggests that linearization stability for gauge-gravity systems should be indexed by a charge fibre—or a stratum of the global bundle moduli—before genericity is discussed. The same linear tangent can have different integrability status in different enlarged families without contradiction.

6. Scope boundary

This is **LOCAL-ALGEBRAIC/REDUCED-MODE**, not a causal nonlinear evolution theorem. The complete harmonic obstruction bilinear, all equal-quantum-number polarization blocks, the full adjoint cokernel, charge-relaxed corrections, and final residual quotient remain open. No asymptotic, scattering, or quantum claim follows. The result is a family of exact obstructions, not yet a theorem that the entire Einstein sector fails nonlinearly.

7. One useful question for adjacent experts

Should the fixed-versus-variable flux distinction be formulated as linearization stability on separate connected components of configuration space, as a stratified moment-map problem, or as an enlarged constraint operator with additional global rows? Which formulation best preserves the classical Taub theorem when the underlying principal bundle changes?

Reproducibility receipt

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source paper: arXiv:1609.07703v1 and its cited classical framework
project certificates: COMPACT_HARMONIC_DOMAIN_AND_TAUB_DESCENT;
EINSTEIN_MAXWELL_OBSTRUCTION_BILINEAR_G1;
EINSTEIN_MAXWELL_WEYL_HERMITIAN_AXIAL_POLAR_ELL2_TAUB
verification: section 3 plus certificate-local commands
dependency tags: LOCAL-ALGEBRAIC; REDUCED-MODE
generality level: G1_DECLARED_COMPACT_HARMONIC_DOMAIN
lifecycle state: DRAFT_ALLOWED
claim flag: CHARGE_FIBRE_DEPENDENT_TAUB_OBSTRUCTIONS_CERTIFIED
open fields: all-harmonic bilinear; full cokernel; charge-relaxed extensions
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